10.1.2 . Reverse a String

A] Algorithm
Step 1: Start
Step 2: Read input string from the user and store it in user_input
Step 3: Define a function reverse_string(input_string)
Step 4: Inside the function, initialize an empty string re = ""
Step 5: Find the length l of input_string
Step 6: Loop from index l-1 down to 0 (reverse order):
    Step 6a: Append the character at current index to re
Step 7: After the loop, return re (the reversed string)
Step 8: Call reverse_string(user_input) and store the result in result
Step 9: Print the original string user_input
Step 10: Print the reversed string result
Step 11: End

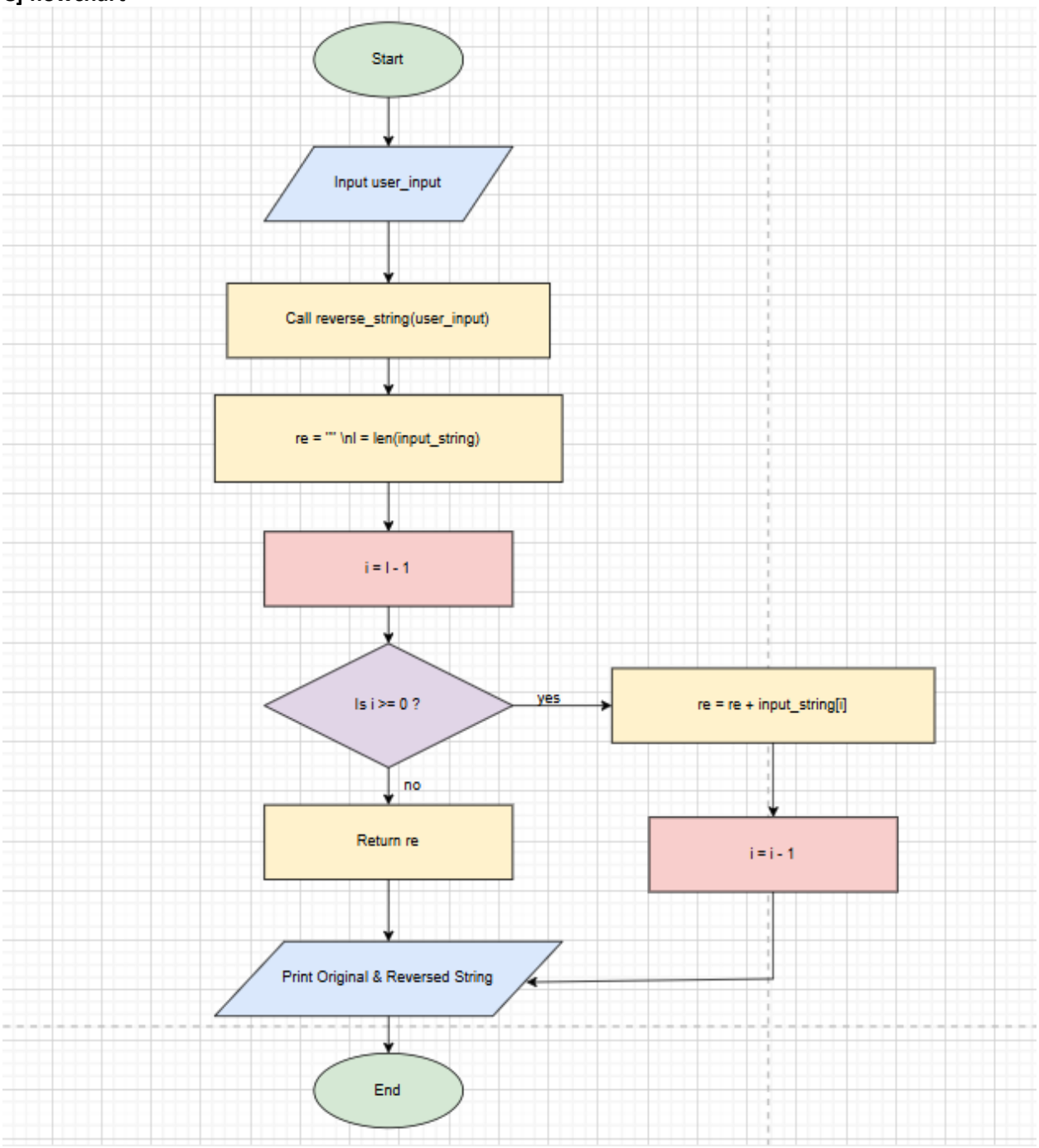
B] code

```
def reverse_string(input_string):
        re=""
        l = len(user_input)
        for i in range(l-1,-1,-1):
                re += user_input[i]
        return re



user_input = input("Enter a string: ")
result = reverse_string(user_input)
print(f"Original String: {user_input}")
print(f"Reversed String: {result}")
```

C] flowchart


Start
Input user_input
Call reverse_string(user_input)
re = "" \nl = len(input_string)
i = l - 1
Is i >= 0 ?
yes
re = re + input_string[i]
no
Return re
i = i - 1
Print Original & Reversed String
End

D] ouput

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10.1.2. Reverse a String 06:58

Write a Python program that prints the reverse of a given input string. The program uses a **reverse_string()** function that takes a string as input and returns its reversed string.

Input Format:

- The program prompts "Enter a string: " and reads a string representing the string to be reversed.

Output Format:

- The first line prints the original string, in the format:

```
Original String: <input_string>
```

- The next line prints the reversed string, in the format:

```
Reversed String: <reversed_string>
```

Sample Test Cases +

ReverseS... Submit

```python
def reverse_string(input_string):
    re=""
    l = len(user_input)
    for i in range(l-1,-1,-1):
        re += user_input[i]
    return re


user_input = input("Enter a string: ")
```

Average time 0.006 s 5.75 ms

Maximum time 0.007 s 7.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 7 ms Debug

Expected output	Actual output
Enter a string: PythonProgrammming	Enter a string: PythonProgrammming
Original String: PythonProgrammming	Original String: PythonProgrammming
Reversed String: gnimmmargorPnohtyP	Reversed String: gnimmmargorPnohtyP

Test case 2 7 ms

Terminal Test cases

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